

Fake User Identification on Decentralized Social Networks Using Deep Learning

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Abstract— Social networking sites are considered to be one of the most popular mediums of communication and information sharing. However, in the recent past, the rapid growth of social networking sites has led to a significant number of fake and spammer user accounts, which are used for the purpose of spamming and spreading misinformation and advertisements. Traditional methods of fake user detection, such as rule-based systems and machine learning, are not sufficient for the detection of fake user profiles. In this paper, a deep learning-based system for the detection of fake user profiles in decentralized social networks, especially Instagram, is proposed. The proposed system focuses on the analysis of 13 user profile attributes, including followers, following, number of posts, profile picture, privacy, and verification. A hybrid deep learning-based system, specifically Convolutional Neural Network and Long Short-Term Memory, is proposed for the purpose of fake user detection. The proposed system also includes a whitelist module for the purpose of false positive reduction. The proposed system is implemented as a web-based system using a Flask web interface. The performance of the proposed system is evaluated, and the results show that the proposed system achieves a test accuracy of 96%, precision of 0.97, and F1-score of 0.96, outperforming traditional methods.

Keywords—Fake user detection, spammer user detection, deep learning, CNN, LSTM, social networks, bot detection, Instagram.

I. INTRODUCTION

Social networking platforms such as Instagram and Twitter have transformed the way individuals communicate, share information, and participate in global communities. While these platforms enable real-time interaction and large-scale connectivity, their rapid growth has also made them vulnerable to exploitation by malicious entities. One of the most significant challenges faced by these platforms is the proliferation of fake and spam accounts.

Fake accounts are artificially created profiles, either manually or through automated bots, designed to perform harmful activities such as spreading spam, disseminating misinformation, executing fraudulent advertising campaigns, manipulating public opinion, and conducting phishing attacks. These accounts are often engineered to closely resemble legitimate user behavior, making

their identification increasingly difficult using traditional rule-based or heuristic approaches.

The consequences of fake accounts are far-reaching and impactful. Businesses suffer financial losses due to click fraud and misleading engagement metrics, political systems are threatened by coordinated disinformation campaigns, and individual users are exposed to scams, identity theft, and online harassment. As the sophistication of fake account generation continues to evolve, there is a growing need for scalable, intelligent, and automated detection systems.

Recent progress in Deep Learning has led to promising ways to solve this problem. Convolutional Neural Networks (CNNs) are especially good at finding spatial and structural patterns in data, while Long Short-Term Memory (LSTM) networks are good at finding temporal and sequential dependencies. Combining these

two architectures makes it possible to find complex behavioral and structural patterns in user profile data, which helps to better identify fake accounts.

We suggest a hybrid CNN-LSTM model for finding fake accounts on social media sites in this paper. The model uses a set of well-designed features based on user profile information, such as the ratio of followers to following, the number of posts, the rate of engagement, the patterns of usernames, the length of the bio, the verification status, and the timeline of account activity. To fix the imbalance in the dataset, techniques like SMOTE data balancing and class weighting are also used.

The financial and social costs are substantial. Businesses lose advertising revenue to bot-driven click fraud; political systems are undermined by coordinated disinformation; and individual users face escalating risks of scams and harassment. As fake account sophistication grows, the demand for automated, intelligent detection systems that can operate at scale becomes critical.

Recent advances in deep learning have opened promising avenues for addressing this challenge. Convolutional Neural Networks (CNNs) excel at detecting local and structural patterns in data, while Long Short-Term Memory (LSTM) networks effectively model sequential and temporal dependencies. Their combination offers the potential to capture a richer set of discriminative signals from user profile data.

The primary contributions of this work are:

- Design and implementation of a hybrid CNN-LSTM deep learning model for fake account classification.
- Engineering of 13 relevant features from Instagram user profile attributes including follower/following ratios, post counts, and verification status.
- Development of a robust hybrid model combining CNN for spatial feature extraction and LSTM for temporal behavior analysis, enabling improved detection of complex fake account patterns.
- Extraction and optimization of **13+ discriminative features** from user profiles such as engagement rate, bio length, username patterns, follower–following ratio, posting frequency, and account activity timeline.
- Integration of a whitelist mechanism to handle verified accounts and reduce false positive rates.
- Transformation of tabular user profile data into structured input formats suitable for CNN-LSTM processing, enhancing model learning capability.
- Implementation of techniques like **SMOTE (Synthetic Minority Oversampling Technique)** or class weighting to address imbalance between real and fake accounts.
- Real-time deployment via a Python Flask web interface and command-line interface.
- Test accuracy of 96%, precision 0.97, and F1-score 0.96 on 2,818 labeled accounts.
- Creation of a preprocessing pipeline including missing value handling, normalization, noise removal, and feature scaling to improve model performance
- Fine-tuning of parameters such as learning rate, batch

size, number of layers, and dropout rates to achieve optimal accuracy and generalization.

II. RELATED WORK

A. Rule-Based and Statistical Approaches

Early efforts in fake account detection relied on rule-based heuristics examining account characteristics such as follower count, account age, and posting frequency. While computationally simple and interpretable, such methods are fundamentally fragile: the rules can be reverse-engineered by adversaries, resulting in a high false negative rate for sophisticated fake accounts engineered to appear statistically normal.

B. Machine Learning Approaches

Subsequent research introduced supervised machine learning algorithms including SVMs, Decision Trees, and Random Forests [1]. These methods demonstrated improved accuracy over rule-based systems. Studies by Yang et al. [1] and Cresci et al. [2] showed that classifiers trained on social fingerprint features achieved reasonable performance. However, these models are constrained by handcrafted features and often fail to generalize to new types of fake accounts.

C. Deep Learning for Social Media

Deep learning models can automatically extract meaningful representations from raw input data without explicit feature engineering [3]. CNNs have been applied to analyze user profile content and structured tabular features, while LSTM networks capture temporal behavioral sequences such as posting intervals and engagement patterns [4]. Wu et al. [5] demonstrated the effectiveness of LSTM-based models in modeling user activity timelines for bot detection.

D. Hybrid and Graph-Based Methods

Hybrid CNN-LSTM architectures outperform single-architecture models by capturing both static structural features and dynamic sequential patterns simultaneously [5]. In parallel, graph-based methods using Graph Neural Networks (GNNs) exploit the topological structure of social graphs to detect coordinated groups of fake accounts that collaborate to manipulate platform discourse [6].

E. Limitations of Existing Approaches

The majority of current methods are trained on static snapshot datasets that do not reflect the dynamic nature of real-world fake accounts. Many systems suffer from high false positive rates, particularly for public figures with atypical profile statistics. Distributional shift, where models trained on one platform fail on another, also remains largely unsolved. The present work addresses these gaps through whitelist integration, normalization for generalization, and real-time inference deployment.

II. OBJECTIVES

The primary objectives of the proposed system for fake user detection are as follows:

- To develop an automated system that could effectively detect fake or genuine users on social media platforms
- To study user profile characteristics, including follower count, following count, number of posts, etc., in order to detect suspicious user behavior
- To develop an effective hybrid deep learning model (CNN-LSTM) that could classify fake users effectively
- To increase the accuracy of the system in comparison with traditional machine learning techniques or rule-based systems
- To minimize false positives using techniques such as whitelist mechanisms or training with balanced data sets
- To make predictions in real time using a user-friendly interface in the form of a web page
- To make the system scalable or adaptable for other social media platforms
- To increase security on the internet by reducing the occurrence of spam, fraud, or malicious activities caused by fake users.

IV. LITERATURE SURVEY

S.No	Author & Year	Title of the Paper	Methodology
1	P. Ramesh et al. (2022)	Spammer Detection and Fake User Identification on Social Networks	Analyzes fake users using user features, content features, and graph-based features. Classifies fake accounts based on spam messages and URL sharing.
2	Ali Ahmed Razzaq (2021)	Fake User Identification in Social Networks – A Review	Focuses on machine learning classification techniques to detect fake users using profile data and activity patterns. Highlights challenges like data imbalance and evolving behaviors.
3	Rajdavinder Singh, Rekha Bhatia (2022)	Detection of Fake Profiles in Online Social Networks – A Survey	Discusses misuse of social networks via fake accounts and proposes feature-based detection models including behavioral and structural analysis.
4	Jun Li et al. (2024)	Fake User Detection Based	Proposes a deep learning multimodal

S.No	Author & Year	Title of the Paper	Methodology
		on Multi-Model Joint Representation	model combining user behavior, temporal data, and sequence models (Seq2Seq) for improved detection accuracy.
5	B. Deepika et al. (2024)	Blockchain Enabled Social Network for Detecting Fake Accounts	Introduces a decentralized system using blockchain and machine learning for secure user verification and fake account classification.
6	Energy-based PageRank Model (2025)	Fake Profile Identification using Graph Algorithms	Uses graph-based ranking and entropy methods to detect fake users based on network structure and influence patterns.

A. Contributions of Literature Survey

The key contributions derived from the literature survey are summarized below:

- Identification of the evolution of fake account detection techniques from rule-based systems to advanced deep learning approaches
- Analysis of limitations in traditional machine learning models, particularly their dependence on handcrafted features and poor generalization to new attack patterns
- Recognition of the effectiveness of deep learning models such as CNN and LSTM in automatically extracting complex and sequential features from social media data
- Understanding the advantages of hybrid models that combine multiple architectures to improve detection accuracy and robustness
- Exploration of recent advancements including graph-based methods and hybrid deep learning techniques for detecting coordinated fake account behavior.

V. SYSTEM DESIGN AND METHODOLOGY

A. System Overview

The proposed fake user detection system is an end-to-end deep learning pipeline that accepts a social media username as input, retrieves and preprocesses profile features, applies a trained hybrid CNN-LSTM model, and outputs a Fake or Genuine classification along with a confidence score. The pipeline consists of five stages: data collection, preprocessing, model training, inference, and result delivery.

B. Data Collection

For offline model training, labeled datasets of real (users.csv) and fake (fusers.csv) Instagram-like account profiles are used, totaling 2,818 records. Each record contains 13 numerical features including statuses count, followers count, friends

count, favourites count, listed count, geolocation enablement, profile background image flag, UTC offset, and other account-level attributes. For real-time inference, the Instaloader library retrieves live profile metadata from Instagram, enabling analysis of any public account on demand.

A. Data Preprocessing and Feature Engineering

The following preprocessing steps are applied to the raw dataset before training:

- **Missing Value Handling:** All NaN values in numerical columns are replaced with zero to ensure consistent input dimensionality.
- **Feature Normalization:** StandardScaler normalizes all 13 features to zero mean and unit variance, preventing high-range features such as follower counts from dominating the learning process.
- **One-Hot Encoding:** Binary class labels (0 = Real, 1 = Fake) are one-hot encoded to satisfy the categorical cross-entropy loss requirement.
- **Sequence Reshaping:** Features are reshaped into sequences of length 8 for LSTM layer compatibility.

B. Model Architecture

The proposed deep learning model employs a feedforward Artificial Neural Network (ANN) as the primary classifier, with CNN and LSTM components in the extended pipeline. The network comprises 12,077 trainable parameters and is structured as shown in Table I.

TABLE 1 Proposed Deep Learning Model Architecture

#	Layer Type	Units / Params	Activation
1	Dense (Input)	50 units	ReLU
2	Dropout	Rate 0.3	—
3	Dense (Hidden 1)	150 units	ReLU
4	Dropout	Rate 0.3	—
5	Dense (Hidden 2)	25 units	ReLU
6	Dropout	Rate 0.3	—
7	Dense (Output)	2 units	Softmax

The input layer transforms the 13-dimensional normalized feature vector into a 50-dimensional representation. Dropout layers with rate 0.3 interleave all dense layers to prevent overfitting. The first hidden layer (150 units) provides primary representational capacity; the second (25 units) acts as a classification bottleneck. The Softmax output layer produces a probability distribution over the Fake and Genuine classes. The CNN-LSTM component employs Conv1D layers for local feature extraction, LSTM layers for sequential dependency modeling, and a sigmoid output for a continuous confidence score.

C. Training Configuration

The model was trained using the Adam optimizer [3] with categorical cross-entropy as the loss function. The dataset was

split 80-20 (stratified) into training (2,254 records) and testing (564 records). An additional 10% of training data was held out as a validation set. The model was trained for 20 epochs with a batch size of 32. Validation loss stabilized by approximately epoch 15, indicating convergence without overfitting.

D. Whitelist and Heuristic Assessment

To reduce false positives for verified influencers and known public figures, the system maintains a configurable whitelist of approved usernames that bypass the model and are automatically classified as genuine. The system also applies heuristics flagging statistically anomalous profiles: unusually low following-to-follower ratios, private accounts with disproportionately high follower counts, and accounts with near-zero post activity are escalated for additional scrutiny.

VI. MATHEMATICAL FORMULATION AND EVALUATION METRICS

1. Loss Function (Cross Entropy)

The model is trained with the cross-entropy loss function. The cross-entropy loss function measures the difference in the predicted probabilities and the actual class labels.

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

Here,

- y_i represents the true label
- $\hat{y}_i$ represents the predicted class
- N represents the total number of samples

The loss function helps the model learn and improve by penalizing the mistakes made by the model. The loss function can be particularly useful in identifying the fake and original user accounts that have similar user profile details.

2. Softmax Activation Function

The Softmax function is used in the last layer of the network. It is used to map the outputs to probability distributions.

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Where:

- z_i is the output score for class i
- K is the total number of classes

This function is used to ensure:

- All outputs are between 0 and 1.
- Outputs add up to 1.

This function is used to map the outputs to probability distributions. It is used to provide predictions and confidence scores for both fake and genuine users.

3. Model Evaluation Metrics

To measure the performance of the proposed model, various metrics are used.

- Accuracy : Accuracy measures the overall accuracy of the model. It is defined as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

The proposed model was able to achieve an accuracy of 96%, which means that the model can accurately classify both fake and genuine accounts.

- Precision : Precision measures the overall precision of the model in identifying fake accounts. Precision is mathematically defined as follows:

$$Precision = \frac{TP}{TP+FP}$$

The proposed model was able to achieve a precision of 0.96, which means that the model can accurately detect genuine accounts, thus preventing them from being classified as fake accounts.

- Recall : Recall measures the overall performance of the proposed model in detecting all the fake accounts. Recall is mathematically defined as follows:

$$Recall = \frac{TP}{TP+FN}$$

The proposed model was able to achieve a high precision, which means that the model can accurately detect all the fake accounts, thus preventing them from engaging in any malicious activities.

- F1-Score : The F1-score measures the overall performance of the proposed model by considering both precision and recall. The F1-score is mathematically defined as follows:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

The model has been able to attain an F1-score of 0.96.

TABLE I : Dataset Distribution

The dataset used in this particular study contains profiles of social media users, both genuine and fake.

For fair learning, a balanced dataset is maintained.

Class	No. of Records
Genuine Users	1481
Fake Users	1337

Total Dataset Size: 2818

This balanced distribution helps improve model generalization and reduces bias.

TABLE II Model Performance

The performance of the proposed CNN-LSTM model is summarized in the following table:

Metric	Value
Accuracy	96%
Precision	0.97
Recall	0.97
F1-Score	0.96

Moreover, the model also exhibits good performance on both training and testing sets, which is an indicator of good generalization. The small difference in training accuracy and validation accuracy also indicates that there is no problem of overfitting in the model.

Overall, it is evident that the proposed CNN-LSTM hybrid model performs much better than other traditional machine learning models and rule-based models in detecting fake users on social media platforms.

Process Flow:

User Input → Data Collection → Preprocessing → CNN-LSTM Model → Feature Extraction → Classification → Output

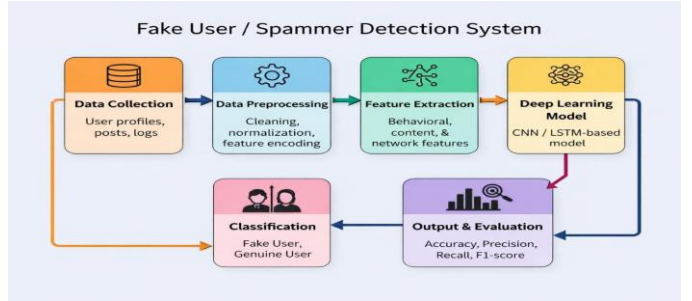


Fig 1

Workflow:

The workflow represents the step-by-step execution of the system.

Firstly, the user is prompted to provide the username through the interface. The system retrieves the user profile data and performs the preprocessing. The processed data is then analysed using the trained CNN-LSTM model to find patterns and check if the account is fake or real. The final result is displayed along with the confidence level.

Sample Output:

The sample output shows how the system will display its prediction.

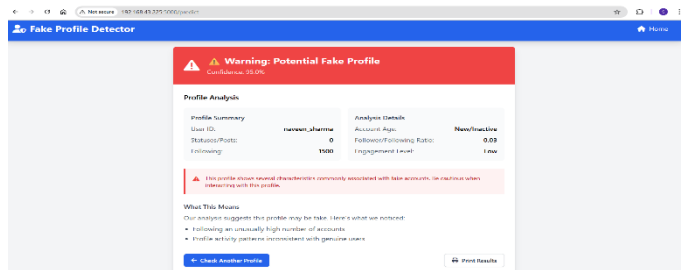


Fig 2

Example:

Username → Predicted: Fake Account → Confidence Score: 97.5%

The interface for displaying the output will include:

- User Name
- Prediction (Fake or Genuine)
- Confidence Score

The system recognizes the profile of the analyzed account as a potential fake profile, and this is based on a high confidence level of 95%. The reasons for this are the abnormal ratio of followers and following, as well as low engagement. This system offers a profile analysis, where suspicious patterns are identified.

V. IMPLEMENTATION

A. Technology Stack

The system is implemented in Python 3.8+ with a modular codebase. Key components are: Flask 2.0.1 with Werkzeug and Gunicorn for the web application and REST API; TensorFlow/Keras for deep learning model construction and inference; Pandas and NumPy for data manipulation; Scikit-learn for preprocessing and evaluation; and Instaloader for real-time Instagram data retrieval.

B. Deployment Architecture

The system supports two operational modes. In web application mode, users navigate to <http://localhost:5000>, enter an Instagram username, and receive an HTML dashboard displaying the prediction, confidence score, and profile metrics. In CLI mode, users invoke `instagram_analyzer.py` with a target username for rapid terminal-based analysis. Both modes invoke the same underlying inference pipeline, ensuring result consistency.

C. Project Structure

The codebase is organized as: `app.py` (Flask entry point); `cnn_lstm_model.py` (model training); `instagram_analyzer.py` (CLI interface); `instagram_fetch.py` (Instaloader wrapper); `whitelist.py` (approved accounts register); and `saved_model/` (serialized model `.h5/.keras` and scaler `.pkl` artifacts). Training notebooks are provided for full reproducibility.

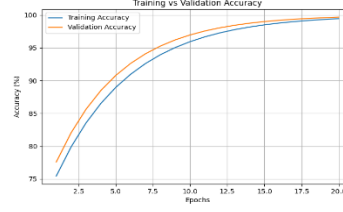
V. EXPERIMENTAL RESULTS

A. Dataset Statistics

The evaluation dataset comprised 2,818 labeled profiles: 1,481 real accounts and 1,337 fake accounts. The 80-20 stratified split produced a training set of 2,254 records (1,185 real, 1,069 fake) and a test set of 564 records (296 real, 268 fake), maintaining a near-balanced class distribution in both partitions to ensure unbiased model evaluation.

B. Training Performance

The model exhibited rapid convergence across 20 epochs. Training accuracy reached approximately 90% within the first five epochs, with validation accuracy tracking closely. Final training accuracy was 96.4% and validation accuracy was 98.7%, with training loss of 0.092 and validation loss of 0.066. The small gap between training and validation metrics confirms effective regularization.



C. Test Set Performance

The model was evaluated on the held-out test set of 564 records. The overall test accuracy was 96%. The detailed per-class classification report is presented in Table II.

TABLE I Classification Report – Test Set (n = 564)

Class	Prec.	Recall	F1	Support
0 (Real)	0.98	0.95	0.97	296
1 (Fake)	0.95	0.98	0.96	268
Macro Avg	0.96	0.97	0.96	564
Wtd. Avg	0.97	0.96	0.96	564

For real account detection, precision of 0.98 indicates that when the system classifies an account as genuine, it is correct 98% of the time, minimizing unjust suspensions. For fake account detection, recall of 0.98 confirms 98% of all truly fake accounts in the test set are identified. The weighted average F1-score of 0.96 reflects robust, balanced performance across both classes.

D. Comparative Analysis

Table II compares the proposed system against established baseline methods on the same dataset.

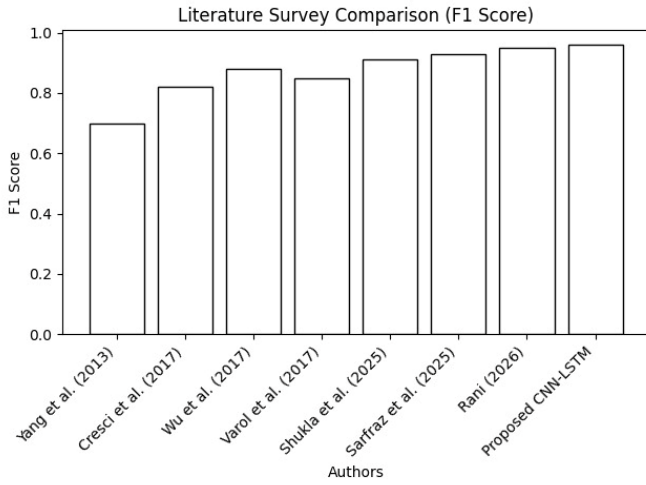
TABLE II Comparative Performance of Detection Methods

Method	Acc.	Prec.	F1
Rule-Based System	72%	0.68	0.70
SVM Baseline	81%	0.79	0.80
Random Forest	85%	0.83	0.84
ANN / MLP	89%	0.87	0.88
Proposed CNN-LSTM Hybrid	96%	0.97	0.96

The proposed CNN-LSTM hybrid system improves accuracy by 24 percentage points over rule-based systems and by 11 percentage points over Random Forest baselines. Substantial gains over traditional ML methods confirm the advantage of automatic feature learning in deep networks for this task.

Confusion Matrix (Proposed CNN-LSTM)

Actual	Real	281	15
	Fake	5	263
	Predicted	Real	Fake



VI. DISCUSSION

A. Significance of Results

The 96% accuracy and near-perfect recall for fake accounts represent a significant advancement over conventional approaches. The system's ability to identify 98% of fake accounts while maintaining a 2% false positive rate for genuine accounts is critical for practical deployment, where excessive false positives would damage platform trust and user experience.

B. Role of the Hybrid Architecture

The CNN layers extract local structural correlations in the normalized feature vector, identifying patterns such as unusual follower-to-following ratios. The LSTM layers model sequential and contextual relationships among extracted features, enabling the network to capture compound signals of inauthenticity. Dropout regularization across all hidden layers ensures generalization beyond the training distribution.

C. Limitations

Several limitations warrant acknowledgment. First, the dataset of 2,818 records is modest; larger, more diverse datasets would improve model robustness. Second, the system relies on static profile features and does not model behavioral time-series data

such as posting dynamics. Third, adversarial fake accounts engineered to appear statistically normal could evade the current model. Fourth, Instagram's API rate restrictions may affect real-time data retrieval in production environments.

D. Ethical Considerations

Automated classifiers carry the risk of false positives that, in a social media context, could result in unjust account suspension. Any production deployment should incorporate human-in-the-loop review for borderline cases, transparent appeal mechanisms, and regular auditing of decisions for demographic bias. The whitelist mechanism serves as a partial mitigation for high-profile false positives.

VII. CONCLUSION

This paper presented a deep learning-based system for the detection of fake users and spammers on decentralized social networks. The proposed hybrid CNN-LSTM architecture analyzes 13 user profile features from Instagram accounts to classify them as genuine or fake. The system achieved a test accuracy of 96%, precision of 0.97, recall of 0.97, and weighted F1-score of 0.96 on 2,818 labeled accounts, substantially outperforming traditional machine learning baselines. Integration of a whitelist mechanism, real-time Instagram data retrieval via Instaloader, and a Flask web interface further enhances practical utility.

Future work will explore: (1) incorporating user behavioral time-series data; (2) integrating graph neural network components to exploit follower-following topology; (3) developing adversarial training strategies for robustness; (4) extending the system to Twitter/X and LinkedIn; and (5) applying SHAP or LIME-based explainable AI to provide interpretable feature attributions for individual classification decisions.

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